
Statistical Methods in Artificial Intelligence

Assignment 3 — Problem Statement

T8.1 · Hindi Product Review Sentiment Analyser

International Institute of Information Technology, Hyderabad
Academic Year 2025–26

Theme	T8 — Indic Review & Sentiment Analyzer
Variant	T8.1 — Hindi Product Review Sentiment
Tier	Tier 1 (1–3 students)
Type	Natural Language Processing (NLP)
Compute	Google Colab Free Tier (T4 GPU) / CPU laptop
Time Budget	4–6 hours per person (total)

1. Problem Overview

India is home to one of the world’s largest and fastest-growing e-commerce ecosystems. Millions of consumers post product reviews every day — a significant proportion of which are written in Hindi, the most widely spoken language in the country. Yet the majority of existing sentiment-analysis tools are designed exclusively for English, leaving Hindi-speaking small business owners and marketers unable to derive structured insights from their customer feedback.

This project addresses that gap by building a **Hindi Product Review Sentiment Analyser**: a machine-learning-powered web application that reads a Hindi product review and classifies it as *Positive*, *Negative*, or *Neutral*, while also reporting a confidence score and surfacing actionable business insights.

2. Motivation & Real-World Impact

- **Language accessibility.** A small Kirana-store owner or a D2C brand selling on Meesho / Flipkart / Amazon India receives the bulk of customer feedback in Hindi. Manually reading and categorising hundreds of reviews is prohibitively time-consuming.
- **Underserved NLP problem.** Multilingual NLP for Indic languages remains an active research frontier; tools like `ai4bharat/indic-bert` have recently made supervised fine-tuning tractable even on small compute budgets.
- **Tangible demo value.** A working web app with an interpretable dashboard constitutes a portfolio-grade project that demonstrates end-to-end ML competency.

3. Objectives

By the end of this project the team will have:

1. Loaded and pre-processed the AI4Bharat **IndicSentiment** Hindi corpus.

- 2. Fine-tuned (or applied zero-shot prompting to) a multilingual language model for three-class sentiment classification.
- 3. Evaluated the model rigorously using accuracy, macro-F1, and a confusion matrix.
- 4. Deployed an interactive Streamlit web application supporting both single-review input and bulk CSV upload.
- 5. Produced a 4–6 page technical report and a one-slide LinkedIn-ready pitch.

4. Dataset

4.1. Primary Dataset

Name	AI4Bharat IndicSentiment (Hindi config)
Source	https://huggingface.co/datasets/ai4bharat/IndicSentiment
Classes	Positive, Negative, Neutral
Scale	~2,000–5,000 labelled examples (Hindi split)
Domain	Product and service reviews in Hindi
Format	HuggingFace <code>datasets</code> arrow format

4.2. Dataset Details

IndicSentiment is a benchmark curated by AI4Bharat covering 13 Indic languages with parallel reviews. The Hindi configuration (`hi`) contains product and service reviews labelled with one of three sentiment classes. The dataset ships with standard train/validation/test splits, making evaluation straightforward and reproducible.

4.3. Optional Data Augmentation

For richer coverage, the team may supplement the primary corpus with:

- LLM-generated synthetic Hindi reviews (via the free Gemini API) labelled by the same LLM and manually spot-checked.
- Publicly available Hindi product reviews scraped (ethically) from open datasets on Kaggle.

5. Proposed Technical Approach

5.1. Modelling Options

Two complementary approaches will be explored and compared:

	Approach A: Fine-tuning	Approach B: Zero-shot
Model	<code>ai4bharat/indic-bert</code> (110M params)	Gemini 1.5 Flash API <i>or</i> Groq Llama-3 API
Training	2–3 epochs, AdamW, cross-entropy loss	No training — structured prompt with JSON output
Inference cost	CPU-friendly after training	API call per review (free tier)
Expected F1	75–85% (benchmarked on IndicSentiment)	70–80% (zero-shot)

The fine-tuned model is the **primary deliverable**; the zero-shot baseline serves as an

ablation comparison point.

5.2. Pipeline

1. **Data loading** — `datasets.load_dataset("ai4bharat/IndicSentiment", "translation-hi")`
2. **Pre-processing** — tokenisation via `AutoTokenizer` (max 128 tokens), label encoding.
3. **Model** — `AutoModelForSequenceClassification` initialised from `indic-bert`, three output logits.
4. **Training** — `Trainer` API or manual PyTorch loop; early stopping on validation macro-F1.
5. **Evaluation** — accuracy, macro-F1, per-class precision/recall, confusion matrix.
6. **Deployment** — export model to `safetensors`; load in Streamlit via `pipeline("text-classification"`

6. Web Application Specification

The Streamlit application must support two interaction modes:

6.1. Single-Review Mode

- Text area accepting Hindi input (Devanagari script).
- On submission: predicted sentiment label, confidence bar, short explanation.

6.2. Bulk CSV Mode

- CSV upload (column: `review_text`).
- Batch inference with progress bar.
- Output: downloadable CSV with sentiment labels and confidence scores.
- Dashboard: pie chart of sentiment distribution, top-5 most positive and most negative review excerpts, keyword word-cloud per sentiment class.

6.3. Deployment Target

HuggingFace Spaces (Streamlit SDK) — free tier. Fallback: Streamlit Community Cloud or a recorded ≤ 2 -minute screen demo.

7. Evaluation Criteria

Metric	Target (Tier 1)	Notes
Test-set accuracy	$\geq 70\%$	Three-class
Test-set macro-F1	≥ 0.68	Balances class imbalance
Inference latency	< 2 s/review	On CPU (Colab or HF Spaces)
App usability	Working demo	Handles empty input gracefully

8. Deliverables

1. **GitHub Repository** (public). Must contain: `README.md`, `requirements.txt`, training notebook (`train.ipynb`), and Streamlit app code (`app.py`).

2. **Live Demo.** Deployed on HuggingFace Spaces or Streamlit Community Cloud. Fallback: recorded screen demo (≤ 2 min).
3. **Technical Report (PDF), 4–6 pages.** Sections: Introduction, Data, Method, Results (with ablation table), Limitations, References, Acknowledgements (LLM usage disclosure).
4. **One-Slide Pitch.** Single PNG/PDF suitable for LinkedIn, summarising the problem, approach, and key result.

9. Technology Stack

Component	Library / Tool
Language model	<code>transformers</code> (HuggingFace), <code>ai4bharat/indic-bert</code>
Training	<code>torch</code> , <code>accelerate</code> , Trainer API
Data	<code>datasets</code> (HuggingFace)
Evaluation	<code>sklearn.metrics</code> , <code>seaborn</code>
Web app	<code>streamlit</code> , <code>plotly</code> / <code>matplotlib</code>
Word-cloud	<code>wordcloud</code>
Zero-shot alt.	Google AI Studio (Gemini 1.5 Flash, free)
Compute	Google Colab T4 GPU (free)

10. LLM Usage Policy

In accordance with assignment guidelines, the team is **encouraged** to use LLMs (Chat-GPT, Claude, Gemini, Copilot) for the following tasks:

- Generating synthetic Hindi review examples for data augmentation.
- Scaffolding Streamlit UI components and dataset loader boilerplate.
- Debugging training loops and resolving library incompatibilities.
- Drafting sections of the technical report (to be reviewed and edited by the team).

Mandatory disclosure: All LLM tools used must be cited in the report’s Acknowledgements section. All evaluation, analysis, and interpretation must be conducted independently by the team, and every line of code must be explainable by the team during the viva.

11. Risk & Mitigation

Risk	Likelihood	Mitigation
HuggingFace dataset unavailable	Low	Mirror to Google Drive before training
GPU quota exhausted on Colab	Medium	Use DistilBERT-multilingual as lighter fallback; or zero-shot path
Class imbalance in dataset	Medium	Weighted loss; report macro-F1
Deployment failures on HF Spaces	Low	Fallback to screen recording

12. References

1. AI4Bharat. *IndicSentiment: A Dataset for Sentiment Analysis in Indic Languages*. <https://huggingface.co/datasets/ai4bharat/IndicSentiment>
2. Kakwani et al. *IndicNLPSuite: Monolingual Corpora, Evaluation Benchmarks and Pre-trained Multilingual Language Models for Indian Languages*. EMNLP Findings, 2020.
3. Devlin et al. *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. NAACL-HLT, 2019.
4. Wolf et al. *Transformers: State-of-the-Art Natural Language Processing*. EMNLP 2020 (Systems Demonstrations).
5. Streamlit Documentation. <https://docs.streamlit.io>
6. HuggingFace Spaces Documentation. <https://huggingface.co/docs/hub/spaces>